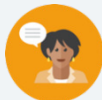


9.5 Solving Quadratics Using Square Roots

Lesson Introduction

 Benchmarks	MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system. MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.		 Learning Targets	- I can solve a quadratic by taking the square root. - I can determine if a solution to a quadratic is non-real. - I can interpret solutions to quadratics given in a real-world context as viable or non-viable.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.		MA.912.AR.3.2 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.	
 Essential Questions	- How are the solutions to quadratic equations different from a linear equation? - How can the solution be found using a graph? - What is a principal square root?			
 Lesson Narrative	In this lesson, students are introduced to solving quadratics using square roots. Students will be guided through an exploration process to reason through the process for solving quadratics using square roots. They will receive formal instruction introducing them to the proper steps to solving and will get guided and independent practice on solving quadratics.			
 Component Summary	9.5.1 Warm-Up (~5 min.)	Relate area of floor tiles to their side length (<i>SE p. 82</i>)	9.5.4 Guided Practice (5 min.)	Practice with solving quadratics by taking the square root (<i>SE p. 87</i>)
	9.5.2 Exploration (15 min.)	Exploration of solving quadratics using square roots (<i>SE p. 82</i>)	9.5.5 Try It (10 min.)	Independent practice with solving quadratics by taking the square root (<i>SE p. 87</i>)
	9.5.3 Guided Instruction (10 min.)	Solving quadratics by taking the square root (<i>SE p. 84</i>)	9.5.6 Wrap-Up (~5 min)	Ticket Out the Door on solving quadratics by taking the square root (<i>SE p. 88</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Principal square root <u>Additional Terms:</u> N/A		N/A	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may only find the principal square root. - Students may not connect the number of solutions with the graphical representation.	- In 9.5.2 Exploration, allow students to explore the concept with a partner without formal definitions and processes. - Students will be engaging with set-builder notation. Spiral any necessary scaffolding without reteaching to maintain the lesson focus on learning targets.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 9.5.3 Guided Instruction question 1 introduces students to the number of solutions that a quadratic equation has by showing a graph. These examples provide an opportunity to poll students to see how many solutions they believe it to have and why.	MA.K12.MTR.2.1: Multiple Representations Demonstrate understanding by representing problems in multiple ways. Students are building connections between the solutions found graphically to the solutions found by finding the square root. This has implications for solving quadratics using other methods.
 Differentiate Instruction	9.5.5 Try It provides students with an opportunity to try to solve quadratic equations using square roots on their own. This section will allow you to determine where students may be struggling in the process to differentiate instruction.	
	The 9.5.6 Wrap-Up activity provides a Ticket Out the Door that allows students to show mastery of the concept of solving quadratics using square roots. The data from this can provide the teacher with data for additional differentiation.	
Lesson Notes:		

9.5.1 Warm-Up: Floor Tiles

Component Overview: Students work independently with tiles to find measurements for each tile. This activity also includes a “discuss with your partner” question where students discuss how the quadratic equation represents the context of the tiles.



9.5.1 Warm-Up: Floor Tiles

Aniyah is renovating her bathroom. She has chosen the floor tile shown. The design uses 4 tiles. The 4 tiles have an area of 2.56 square feet (ft^2). Aniyah wants to determine the measurements, in feet, for each tile.



- Determine the measurements, in feet, for each tile.

$$4x^2 = 2.56$$

$$x^2 = 0.64$$

$$x = \pm 0.8$$

Each tile is a 0.8 by 0.8 ft tile.

- Discuss with your partner how the equation $4x^2 = 2.56$ represents the context. Summarize the discussion.

Answers will vary. The 4 represents the four tiles, x^2 represents the formula for the area of one tile, and 2.56 represents the total area of all four tiles.



Students may use a variety of methods to determine the measurement of each tile. Students are not expected to use an equation but to use a geometric understanding of squares.



Question 2 can act as a transition to lesson content. Students will explore these questions in the next component, so it is not necessary to answer all at this point in the lesson.

- What does $4x^2$ represent?
- Why is the coefficient of x^2 4?
- What would x^2 represent?
- What is the area of one square?



Have students share out their answer to question 2 to allow students to hear multiple perspectives. Consider polling the students to see who has the same perspective and if there's any feedback from their peers.

9.5.2 Exploration: Solving Quadratics Using Square Roots

Component Overview: Students explore the concept of solving quadratics using square roots. This is an Exploration activity, requiring students to work with a partner or small group. For questions 1 and 2, allow students to work collaboratively and then come back together to review answers and student thinking before students continue exploring in questions 3-6.



9.5.2 Exploration: Solving Quadratics Using Square Roots

1. The steps for solving $4x^2 = 2.56$ are shown. Complete the table by completing the explanation for the tile context.

Step	Tile Context
$4x^2 = 2.56$	Four square tiles with area x^2 have an area of 2.56 ft^2
$\frac{4x^2}{4} = \frac{2.56}{4}$ Divide both sides by 4 $x^2 = 0.64$	The equation $x^2 = 0.64$ represents <i>the area of one tile.</i>
$\sqrt{x^2} = \sqrt{0.64}$ Take the square root of both sides $x = 0.8$ and $x = -0.8$	The values $x = 0.8$ and $x = -0.8$ represent the tile's <i>length.</i>
$x = 0.8$	One tile in the pattern has a length of 0.8 feet. A negative 0.8 feet does not make sense for this context.

2. How do the steps in the table relate to the steps you used in the Warm-Up to determine the measurements, in feet, for each tile?
Answers will vary. I knew the area formula needed and worked backwards to figure it out.
3. Wyatt wondered if he could solve $4x^2 = 2.56$ by first taking the square root of both sides. His work is shown.

$$\begin{aligned}\sqrt{4x^2} &= \sqrt{2.56} \\ 2x &= 1.6 \\ x &= 0.8\end{aligned}$$

Wyatt's teacher told him that he could solve this quadratic by taking the square root of both sides first, but that he made a mistake in the second step of his work. Correct Wyatt's work.

$$\begin{aligned}\sqrt{4x^2} &= \pm\sqrt{2.56} \\ 2x &= 1.6 & 2x &= -1.6 \\ x &= 0.8 & x &= -0.8\end{aligned}$$



- Why do you think there would be two answers when taking the square root of both sides?
- What is the value of $(0.8)^2$? $(-0.8)^2$?



While circulating, ask students to share their answers to question 2. Allow them to engage in dialogue on how they believe the table relates to the steps they used in the Warm-Up.

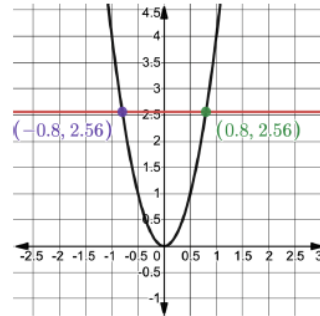
Students may note that they divided by 4, then found the square root. If the student only gives a positive for the square root, ask what they remember about the principal square root. This is addressed later in the lesson.



Challenge students to explain why taking the square root of both sides could work as the first step.

9.5.2 Exploration: Solving Quadratics Using Square Roots (continued)

4. Dominic wanted to find the solution to $4x^2 = 2.56$ using a graphing tool. He graphed $y = 4x^2$ and $y = 2.56$. His graph is shown.



Discuss with your partner how the graph shows the solutions to $4x^2 = 2.56$. Summarize your discussion.

Answers will vary. The horizontal line intersects the parabola at two points, where $x = -0.8$ and $x = 0.8$.

5. Takeisha created a table of values for $y = 4x^2$.

x	-1	-0.9	-0.8	-0.7	-0.6	-0.5	-0.4	-0.3	-0.2	-0.1	0
y	4	3.24	2.56	1.96	1.44	1	0.64	0.36	0.16	0.04	0
x	1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	
y	4	3.24	2.56	1.96	1.44	1	0.64	0.36	0.16	0.04	

- Work with your partner to find the missing values in the table.
Answers in table above.
 - Describe the process you and your partner used to find the missing values.
Answers will vary. My partner and I found x -values that were opposites and used the associated y -values.
 - How can Takeisha use the table of values to solve $4x^2 = 2.56$?
Answers will vary. She can use the table to find where the y -value is 2.56 to find the corresponding values of x .
6. Out of all of these representations, the answer $4x^2 = 2.56$ that Aniyah should use is 0.8 ft because the length must be positive, and the total area must be 2.56 square feet.



What do you notice about the solution shown on the graph and the answer you found in the 9.5.1 Warm-Up?



While circulating the room, listen to students as they discuss the process used to find the missing values. Take notes on any misconceptions that could be used for the routine of "my favorite mistake."



Challenge students to come up with two additional ways Takeisha could solve the problem without using the table of values.

9.5.3 Guided Instruction: Solving Quadratics by Taking the Square Root

Component Overview: Students receive guided instruction on solving quadratics by taking the square root. Begin this activity by going over the content prior to question 1, highlighting key vocabulary such as “principal square root” and review of the content written in set-builder notation.



9.5.3 Guided Instruction: Solving Quadratics by Taking the Square Root

The square root of a number is the value that, when multiplied by itself, results in the radicand. For example, $\sqrt{25}$ is 5 or -5 .



The **principal square root** is the positive square root of a positive real number.

The principal square root is shown algebraically using $\sqrt{a^2} = |a|$ where $\{a | a \in R\}$. If $a < 0$ then $|a|$ is positive and if $a > 0$, $|a|$ is also positive.

In the Exploration, the steps to solving the equation $4x^2 = 2.56$ were shown.

The step $\sqrt{x^2} = \sqrt{0.64}$ resulted in $x = 0.8$ and $x = -0.8$. The steps below can be used to show this algebraically using the relationship $\sqrt{a^2} = |a|$ where $\{a | a \in R\}$.

$$\sqrt{x^2} = \sqrt{0.64}$$

$$|x| = 0.8$$

$$x = 0.8 \text{ and } x = -0.8$$

Because $\sqrt{a^2} = |a|$ for all real numbers, $\sqrt{x^2} = |x|$.

Since $|0.8| = 0.8$ and $|-0.8| = 0.8$ then the x in $|x|$ could be 0.8 or -0.8 .

Unless otherwise stated, the “square root” of a number refers **only** to the principal square root. When solving $x^2 = a$ using square roots, the answer should **always** be both the positive and negative values.



Use your observations during the 9.5.2 Exploration to address misconceptions during the 9.5.3 Guided Instruction.

Review the difference between the terms “square root” and “principal square root.” Consider using a vocabulary method such as the Frayer model for the term “principal square root.” This can help students connect to that vocabulary and can also assist ELL students with this concept.



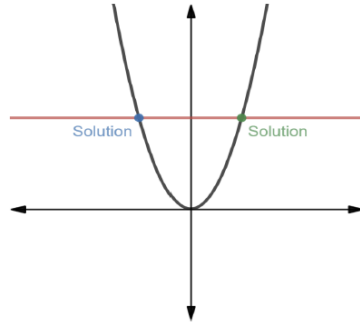
How would the set-builder notation be read out loud?



Consider having students summarize what was addressed on this page. Have them tell a partner what they heard and then switch. Call on a few sample pairs to hear their perception of what was covered.

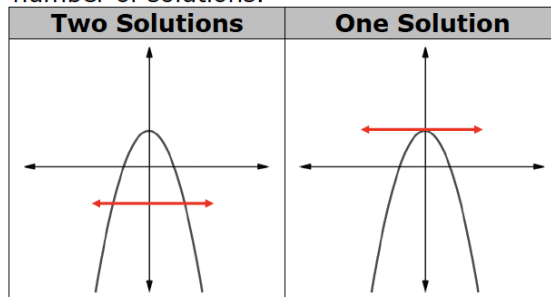
9.5.3 Guided Instruction: Solving Quadratics by Taking the Square Root (continued)

The graph demonstrates why there are two solutions when solving a quadratic.

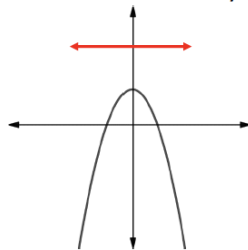


1. Work with your partner to answer the following questions.

- a. Draw a horizontal line for $y = a$ to demonstrate the number of solutions.



- b. Describe where the horizontal line intersects with the parabola when the quadratic has one solution.
Answers will vary. The horizontal line intersects the parabola at the vertex.
- c. Discuss with your partner where a horizontal line could be drawn so that it does not intersect with the parabola. Demonstrate this by drawing a horizontal line.



Why does the place in which the horizontal line intersects the quadratic represent the solution?



Consider creating a visual representation of this information, such as an anchor chart that shows when the answer will be which place. Alternatively, create a solution foldable to show students what no solution, one solution, or two solutions look like on a graph of a parabola.



When there are two solutions present, do you think they will be equidistant? Why or why not?



What does it mean when the horizontal line does not intersect the parabola at all?

9.5.3 Guided Instruction: Solving Quadratics by Taking the Square Root (continued)

2. Consider the equation $x^2 + 2 = 2$.

a. Find the solution to $x^2 + 2 = 2$.

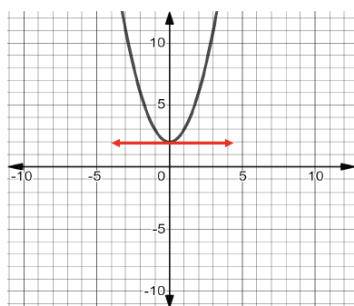
$$x^2 = 2 - 2$$

$$x^2 = 0$$

$$\sqrt{x^2} = \sqrt{0}$$

$$x = 0$$

b. The graph of $y = x^2 + 2$ is shown. Draw a horizontal line that shows $y = 2$.



c. Explain how the location of the intersection of the horizontal line and the parabola shows that $x^2 + 2 = 2$ only has one solution.

Answers will vary. The horizontal line only touches the parabola at one point, the vertex, causing the quadratic equation to have only one solution.



Ask students to relate the graph of $y = x^2 + 2$ and $y = 2$ to the graph of $y = x^2$ and $y = 0$. Ask students how translating $y = x^2 + 2$ to $y = x^2$ is the same as translating $y = 2$ to $y = 0$.



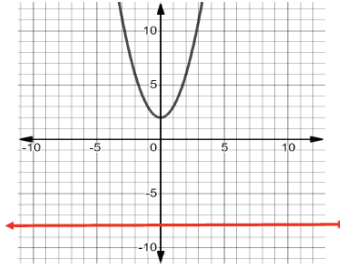
Why does the line not cross or touch the parabola when there is no solution? Describe if one solution always occurs at the vertex.



Have students share out a recap of the different types of solutions based on the graph. Consider sketching a parabola and drawing a horizontal line through it and having students determine how many solutions are shown.

9.5.3 Guided Instruction: Solving Quadratics by Taking the Square Root (continued)

3. The graph of $y = x^2 + 2$ is shown.



- a. Draw the horizontal line $y = -7$.

Answer on graph above.

- b. Find the solution to $x^2 + 2 = -7$.

$$x^2 + 2 = -7$$

$$x^2 = -7 - 2$$

$$x^2 = -9$$

The square root of a negative number is an imaginary number, therefore there is no real solution.

- When solving $x^2 = a$, where $\{a: a > 0\}$ the answer will always be both the positive and negative values of $\sqrt{a}$.
- When solving $x^2 = a$, where $\{a: a = 0\}$ the answer will be 0.
- When solving $x^2 = a$, where $\{a: a < 0\}$ the answer is non-real.

The topic of non-real numbers (imaginary numbers) will be learned in Algebra 2.



For students moving at a strong pace and understanding, consider having them do a teach-aloud where they teach their partners how to determine the number of solutions.



Be explicit and clear with the language of these examples. Refer to what has been addressed throughout this entire activity, referencing the interaction between the horizontal line and the graph of the parabola. Help students understand the algebraic representation by asking for a value for a that fits the set of values represented by the set-builder.



What kind of solution would exist when you take the square root of a negative number?



Consider connecting a visual representation, such as an anchor chart, of the three types of solutions with the algebraic representation of the three types of solutions.

9.5.4 Guided Practice: Solving Quadratics by Taking the Square Root

Component Overview: Students will engage in a guided practice experience with solving quadratics by taking the square root. This activity has only one question. Review the problem with students and release them to complete each step.



9.5.4 Guided Practice: Solving Quadratics by Taking the Square Root

Solve the equation. Show your work.

$$\begin{aligned} 1. \quad x^2 - 41 &= -5 \\ x^2 &= -5 + 41 \\ x^2 &= 36 \\ \sqrt{x^2} &= \pm\sqrt{36} \\ x &= 6; x = -6 \end{aligned}$$



Consider combining this question with Try It questions.



How can we tell if the quadratic equation will result in two unique answers?

9.5.5 Try It: Solving Quadratics by Taking the Square Root

Component Overview: Students will work independently on solving quadratics by taking the square root. This activity is designed to be done independently or can be done in pairs or small groups. Since this lesson has a Ticket Out the Door, consider allowing partner work for students to collaborate on the questions.



9.5.5 Try It: Solving Quadratics by Taking the Square Root

Solve each equation. Show your work.

1. $-5x^2 = -180$

$$x^2 = -\frac{180}{-5}$$

$$x^2 = 36$$

$$\sqrt{x^2} = \pm\sqrt{36}$$

$$x = 6; x = -6$$

2. $39 = 3(x^2 + 7)$

$$39 = 3(x^2 + 7)$$

$$\frac{39}{3} = x^2 + 7$$

$$13 = x^2 + 7$$

$$13 - 7 = x^2$$

$$6 = x^2$$

$$\pm\sqrt{6} = \sqrt{x^2}$$

$$x = \sqrt{6}; x = -\sqrt{6}$$



For students who finish early, challenge them to create a quadratic equation that could be solved using the square root method and have them swap with their partner to solve.



Question 2 may not “look familiar” compared to other quadratic equations. For students who are unsure how to begin, consider reorganizing the equation and having students explain what they notice.

9.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that allows them to show mastery of solving quadratics by taking the square root. This activity should be done independently so that data can be collected on student understanding of the lesson content.



9.5.6 Wrap-Up: Ticket Out the Door

- Find the solution(s). Show your work.

$$86 = 5x^2 - 34$$

$$86 + 34 = 5x^2$$

$$120 = 5x^2$$

$$\frac{120}{5} = x^2$$

$$24 = x^2$$

$$\pm\sqrt{24} = x$$

$$x = \sqrt{24}; x = -\sqrt{24}$$

$$x = 2\sqrt{6}; x = -2\sqrt{6}$$



Consider using the Ticket Out the Door as a formative assessment to gather data on student progress from this lesson.